NanoGT Match Report: CDKL5 deficiency disorder

Disease: CDKL5 deficiency disorder (ORPHA:163934)
Primary gene: CDKL5
Gene CDS: 3093 bp
Inheritance: X-linked dominant
Target tissues scored: CNS

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: Vector does not naturally cover all annotated disease tissues: cns; Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable; Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition.

Disease Mechanism Evidence

Molecular mechanism: haploinsufficiency
Mechanistic detail: X-linked haploinsufficiency of CDKL5 serine/threonine kinase disrupts neuronal cytoskeletal and synaptic signalling cascades causing early-onset refractory seizures and severe intellectual disability
Gene-addition compatibility: conditional
Preferred modality class: aav9 cns cdkl5 gene addition
Evidence level/status: direct / source_checked
Evidence summary: Van Bergen NJ et al. (2022 Biochem Soc Trans PMID 35997111) reviewed CDKL5 molecular pathogenicity and the rationale for AAV9-CDKL5 gene therapy; gene addition is conditional because CDKL5 exhibits dose- and isoform-dependent substrate specificity — expression must match endogenous neuronal levels
Evidence source: Van Bergen NJ et al. 2022 Biochem Soc Trans PMID 35997111

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Skysona	LV	7.2/10	Medium	approved
2	Libmeldy	LV	7.0/10	Medium	approved
3	OAV101-IT	AAV9	7.0/10	Medium	approved

Rank	Program	Vector	Score	Confidence	Approval
4	Zolgensma	AAV9	7.0/10	Medium	approved
5	Strimvelis	LV	7.0/10	Medium	approved

Match #1: Skysona

Precedent disease: Cerebral adrenoleukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 7.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.19**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 3093bp / cargo 8000bp (39% utilized)
- Precedent target match: cns
- Protein class mismatch

- Inheritance match (X-linked dominant <-> XL)
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — X-linked haploinsufficiency of CDKL5 serine/threonine kinase disrupts neuronal cytoskeletal and synaptic signalling cascades causing early-onset refractory seizures and severe intellectual disability
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Van Bergen NJ et al. (2022 Biochem Soc Trans PMID 35997111) reviewed CDKL5 molecular pathogenicity and the rationale for AAV9-CDKL5 gene therapy; gene addition is conditional because CDKL5 exhibits dose- and isoform-dependent substrate specificity — expression must match endogenous neuronal levels
- Mechanism source: Van Bergen NJ et al. 2022 Biochem Soc Trans PMID 35997111 (https://pubmed.ncbi.nlm.nih
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — CDKL5 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cns
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #2: Libmeldy

Precedent disease: Metachromatic leukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.05**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 3093bp / cargo 8000bp (39% utilized)
- Precedent target match: cns
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — X-linked haploinsufficiency of CDKL5 serine/threonine kinase disrupts neuronal cytoskeletal and synaptic signalling cascades causing early-onset refractory seizures and severe intellectual disability
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Van Bergen NJ et al. (2022 Biochem Soc Trans PMID 35997111) reviewed CDKL5 molecular pathogenicity and the rationale for AAV9-CDKL5 gene therapy; gene addition is conditional because CDKL5 exhibits dose- and isoform-dependent substrate specificity — expression must match endogenous neuronal levels
- Mechanism source: Van Bergen NJ et al. 2022 Biochem Soc Trans PMID 35997111 (https://pubmed.ncbi.nlm.nih
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more

targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — CDKL5 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cns
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #3: OAV101-IT

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/spinal cord
Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue

Dimension	Score	Max	What it measures
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.05**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 3093bp / cargo 4700bp (66% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — X-linked haploinsufficiency of CDKL5 serine/threonine kinase disrupts neuronal cytoskeletal and synaptic signalling cascades causing early-onset refractory seizures and severe intellectual disability
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Van Bergen NJ et al. (2022 Biochem Soc Trans PMID 35997111) reviewed CDKL5 molecular pathogenicity and the rationale for AAV9-CDKL5 gene therapy; gene addition is conditional because CDKL5 exhibits dose- and isoform-dependent substrate specificity — expression must match endogenous neuronal levels
- Mechanism source: Van Bergen NJ et al. 2022 Biochem Soc Trans PMID 35997111 (https://pubmed.ncbi.nlm.nih
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — CDKL5 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed

instead of simple addition
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Zolgensma

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/motor neuron
Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.05**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 3093bp / cargo 4700bp (66% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score

- Disease mechanism: haploinsufficiency — X-linked haploinsufficiency of CDKL5 serine/threonine kinase disrupts neuronal cytoskeletal and synaptic signalling cascades causing early-onset refractory seizures and severe intellectual disability
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Van Bergen NJ et al. (2022 Biochem Soc Trans PMID 35997111) reviewed CDKL5 molecular pathogenicity and the rationale for AAV9-CDKL5 gene therapy; gene addition is conditional because CDKL5 exhibits dose- and isoform-dependent substrate specificity — expression must match endogenous neuronal levels
- Mechanism source: Van Bergen NJ et al. 2022 Biochem Soc Trans PMID 35997111 (https://pubmed.ncbi.nlm.nih
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — CDKL5 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: Strimvelis

Precedent disease: ADA-SCID
Vector: LV
Tissue target: hematopoietic
Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**6.95**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 3093bp / cargo 8000bp (39% utilized)
- No tissue overlap (disease: [‘CNS’], vector: [‘hematopoietic’])
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — X-linked haploinsufficiency of CDKL5 serine/threonine kinase disrupts neuronal cytoskeletal and synaptic signalling cascades causing early-onset refractory seizures and severe intellectual disability
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Van Bergen NJ et al. (2022 Biochem Soc Trans PMID 35997111) reviewed CDKL5 molecular pathogenicity and the rationale for AAV9-CDKL5 gene therapy; gene addition is conditional because CDKL5 exhibits dose- and isoform-dependent substrate specificity — expression must match endogenous neuronal levels
- Mechanism source: Van Bergen NJ et al. 2022 Biochem Soc Trans PMID 35997111 (https://pubmed.ncbi.nlm.nih
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more

targeted dose

- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — CDKL5 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cns
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition